How many different materials are there?

1 Choose a material from the list that has the property needed.

copper	cotton	glass	rubber	sand	steel

a You can see through it. _____

b It can be easily stretched. _____

c It can conduct electricity. _____

d It can be poured from one container to another. _____

e It can be made into a fabric suitable for clothes. _____

f It can be made into a magnet. _____

2 Approximately how many elements are known today?
Put a ring around the best answer.

10 50 100 500

3 Here are some materials used to make clothes.
Complete the table showing whether these materials occur naturally or whether they are made by people in a factory.

cotton	silk	polyester	PVC	rayon	wool

One has been done for you.

Natural	Factory made
	rayon

4 Wood and polyester are both made up of three elements.
These elements have symbols C, H and O.
Write down the three elements that are present in wood and polyester.

_____ _____ _____

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What are elements made from?

1 Complete the sentence.

Elements are made up of one type of a _ _ _.

2 Here are some **symbols** for some elements. Write the correct symbol beside each element below. The Periodic Table below will help you.

B	Br	C	Ca	Cl	Co	Cr	Cu	F	Fe	Hg	I	N	Ni	P

nitrogen _____ iodine _____ boron _____

carbon _____ phosphorus _____ fluorine _____

nickel _____ cobalt _____ calcium _____

bromine _____ chlorine _____ chromium _____

copper _____ mercury _____ iron _____

3 Complete the sentence.

In question 2, two elements do not have the same first letter as their symbols.

These two elements are _____ and _____.

4 Complete the sentence.

A new material has been made by scientists in which all of the atoms are the same.

This material is a new e _ _ _ _ _ _.

			H hydrogen														He helium
Li lithium	Be beryllium											B boron	C carbon	N nitrogen	O oxygen	F fluorine	Ne neon
Na sodium	Mg magnesium											Al aluminium	Si silicon	P phosphorus	S sulphur	Cl chlorine	Ar argon
K potassium	Ca calcium	Sc scandium	Ti titanium	V vanadium	Cr chromium	Mn manganese	Fe iron	Co cobalt	Ni nickel	Cu copper	Zn zinc	Ga gallium	Ge germanium	As arsenic	Se selenium	Br bromine	Kr krypton
Rb rubidium	Sr strontium	Y yttrium	Zr zirconium	Nb niobium	Mo molybdenum	Tc technetium	Ru ruthenium	Rh rhodium	Pd palladium	Ag silver	Cd cadmium	In indium	Sn tin	Sb antimony	Te tellurium	I iodine	Xe xenon
Cs caesium	Ba barium	La lanthanum	Hf hafnium	Ta tantalum	W tungsten	Re rhenium	Os osmium	Ir iridium	Pt platinum	Au gold	Hg mercury	Tl thallium	Pb lead	Bi bismuth	Po polonium	At astatine	Rn radon
Fr francium	Ra radium	Ac actinium															

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What are elements like?

Here is the Periodic Table. It may help you answer these questions.

							H hydrogen										He helium
Li lithium	Be beryllium											B boron	C carbon	N nitrogen	O oxygen	F fluorine	Ne neon
Na sodium	Mg magnesium											Al aluminium	Si silicon	P phosphorus	S sulphur	Cl chlorine	Ar argon
K potassium	Ca calcium	Sc scandium	Ti titanium	V vanadium	Cr chromium	Mn manganese	Fe iron	Co cobalt	Ni nickel	Cu copper	Zn zinc	Ga gallium	Ge germanium	As arsenic	Se selenium	Br bromine	Kr krypton
Rb rubidium	Sr strontium	Y yttrium	Zr zirconium	Nb niobium	Mo molybdenum	Tc technetium	Ru ruthenium	Rh rhodium	Pd palladium	Ag silver	Cd cadmium	In indium	Sn tin	Sb antimony	Te tellurium	I iodine	Xe xenon
Cs caesium	Ba barium	La lanthanum	Hf hafnium	Ta tantalum	W tungsten	Re rhenium	Os osmium	Ir iridium	Pt platinum	Au gold	Hg mercury	Tl thallium	Pb lead	Bi bismuth	Po polonium	At astatine	Rn radon
Fr francium	Ra radium	Ac actinium															

1 Use words from this list to complete the sentences.

oxygen	**hydrogen**	**metals**	**non-metals**

a Elements on the left-hand side of the Periodic Table are _____.

b Elements on the right-hand side of the Periodic Table are _____.

c The first element in the Periodic Table is _____.

d The element to the right of nitrogen in the Periodic Table is _____.

2 Put a tick in the box if the element is a metal. Put a cross if it is a non-metal.

☐ fluorine

☐ sodium

☐ neon

☐ lithium

3 Use elements from this list to answer the questions.

aluminium	**bromine**	**chlorine**

a Which element is a gas at room temperature? _____

b Which element is liquid at room temperature? _____

c Which element has many different properties from the other two? _____

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How do we get all the other materials?

1 Use words from this list to complete the sentences.

atom	element	compound	molecule

a A substance made up of molecules in which all the atoms are the same is an _____.

b A substance made up of molecules in which there are atoms of different elements combined is a _____.

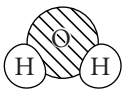
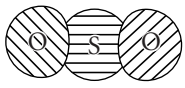
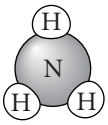
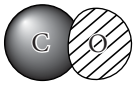
c The smallest part of an element that can exist is called an _____.

d The smallest part of a compound that can exist is called a _____.

e A substance made up of different elements combined in fixed proportions is a _____

f A substance made up of one type of atom is an _____.

2 Complete the table that shows the elements present in different compounds.

Compound	**Elements present**
water 	hydrogen and oxygen
_____	copper and oxygen
sulphur dioxide 	_____ and _____
ammonia 	_____ and _____
carbon monoxide 	_____ and _____

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How can we record the changes when atoms join?

1 Word equations can be used to show chemical reactions.

When hydrogen burns in oxygen, water is produced.

The word equation is:

hydrogen + oxygen → water

a Write the word equation for iron and sulphur combining to form iron sulphide.

_____

b Write a word equation for the burning of magnesium in oxygen.

_____

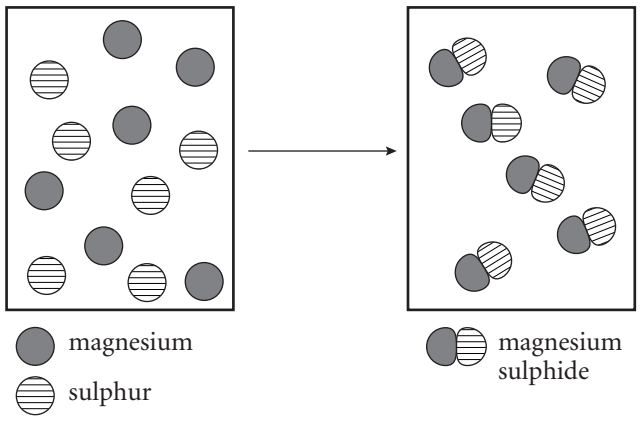
c The word equation for the reaction of calcium and oxygen is:

calcium + oxygen → calcium oxide

What are the names of the two substances that react together?

_____

2 The diagram shows how magnesium and sulphur atoms change as they react.



a Write a word equation for this reaction.

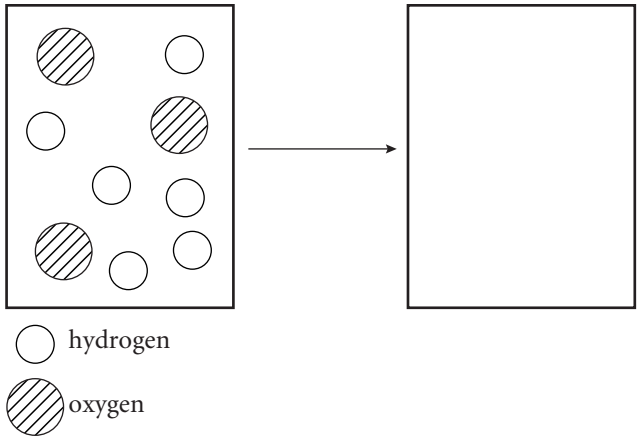
_____

b What happens to the atoms during the reaction?

_____

3 Hydrogen and oxygen atoms combine together to form water molecules. Each water molecule is made up of two hydrogen atoms and one oxygen atom.

Complete the diagram.



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Getting ahead of the game

1 Complete the table that shows the elements present in different compounds.

Compound	**Elements present**
copper oxide	copper and oxygen
copper chloride	_____ and _____
copper sulphide	_____ and _____
copper nitrate	_____, _____ and _____
sodium oxide	_____ and _____
sodium sulphate	_____, _____ and _____

2 The following things were seen when lead was heated in air.

The lead melted and formed a silvery liquid.

A yellow powder called lead oxide formed on the surface of the liquid.

a What gas in the air reacts with lead? _____

b What elements are in the yellow powder?

_____ and _____

c Complete the word equation for the reaction:

_____ + _____ → lead oxide

3 Here is a word equation:

sodium + chlorine → sodium chloride

a What substances are the reactants?

_____ and _____

b What substance is the product? _____

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All about elements and compounds

1 Use words from this list to complete the sentences.
You may use each word more than once.

atom	compound	atoms	element	formula	molecule	elements

An _____ is a substance made up of only one type of atom.

An _____ is the smallest part of an element that can exist.

A group of atoms joined together in an element or a compound is called a _____.

Hydrogen and oxygen are _____.

Hydrogen and oxygen react together to form a _____ called water.

Water has the _____ H_2O.

Writing H_2O means that two hydrogen _____ are joined to one oxygen atom to make one water _____.

MgO is the _____ for magnesium oxide. It shows that there are equal numbers of magnesium and oxygen _____ in magnesium oxide.

$ZnCl_2$ is the _____ for zinc chloride. It shows that for every zinc atom, there are two chlorine _____.

2 The table shows some formulae and what each one means.
Complete the table. One has been done for you.

Formula	**Meaning**
H	an atom of hydrogen
N	
H_2	
N_2	
NH_3	a _____ of ammonia

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How do compounds differ from the elements in them?

1 a Ammonia can be made from nitrogen and hydrogen. Complete the word equation.

nitrogen + _____ → _____

The table gives some properties of nitrogen, hydrogen and ammonia.

	Nitrogen	**Hydrogen**	**Ammonia**
State at room temperature	gas	gas	gas
Changes on cooling to −40 °C	no change	no change	turns to liquid
With a lighted splint	goes out	burns with a squeaky pop	goes out
With a piece of damp red litmus paper	no change	no change	turns blue
Add some water	no change	no change	all the gas dissolves in water

A mixture of elements has the same properties as the elements in the mixture.

b You have a jar containing a mixture of nitrogen and hydrogen.
What would you expect to happen if:

the gas in the jar was cooled to −40 °C?

_____

a piece of damp red litmus paper was added?

_____

some water was added?

_____

c You have a jar of ammonia gas. What would you expect to happen if:

the gas in the jar was cooled to −40 °C?

_____

a lighted splint was put into the gas?

_____

a piece of damp red litmus paper was added?

_____

some water was added?

_____

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Do compounds react chemically?

1 Sam and Zoe heated some zinc oxide in a test tube. They then let it cool again. They weighed the test tube and zinc oxide before and after heating. Their observations and results are shown below.

Zinc oxide is a white powder.
When it is heated it turns to a yellow powder without melting.
On cooling it turns back to a white powder.
Mass of test tube and contents before heating = 40.57 g
Mass of test tube and contents after heating = 40.57 g

Sam says that a chemical reaction has taken place because the zinc oxide changed colour. Zoe says there is no chemical reaction.

a Who do you think is right? _____

b Suggest two reasons why Zoe thinks there is no chemical reaction.

i _____

ii _____

2 For each of the following, write 'yes' if there is a chemical reaction and 'no' if there is not.

a melting ice _____

b burning a piece of wood _____

c dissolving sugar in water _____

d freezing water _____

e a piece of steel rusting _____

f a mixture of hydrogen and oxygen exploding with a squeaky pop _____

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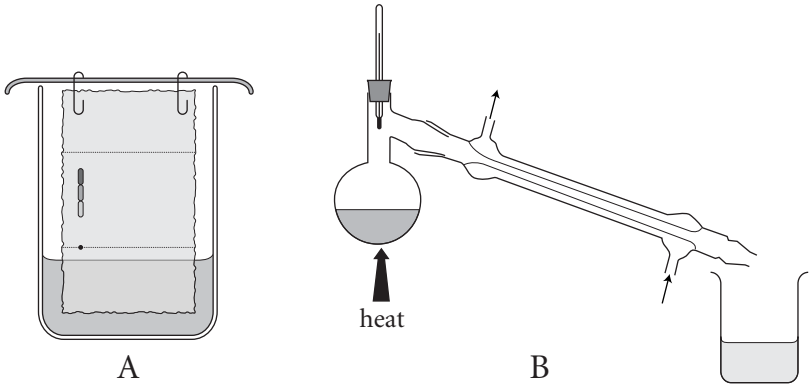
Compounds and mixtures

1 Complete the table by writing each substance below in the correct column. One has been done for you.

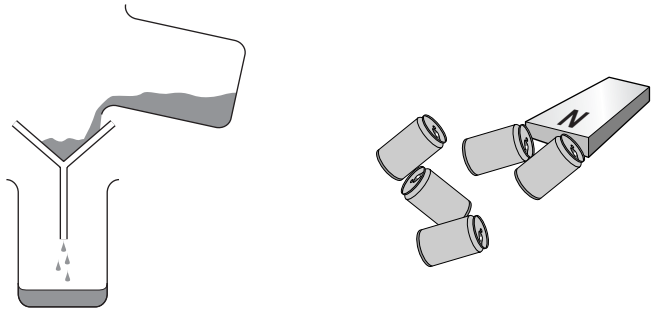
air **blood** **milk** **methane** **pure water** **salt** **sea water** **carbon dioxide**

Single compound	**Mixture**
	milk

2 The diagrams show four ways of separating mixtures.



A B



C D

Which method can be used for separating each of the following?

a pieces of glass from a solution _____

b steel cans from a mixture of steel and aluminium cans _____

c different dyes in the ink from a felt pen _____

d sand from a mixture of sand and sea water _____

e pure water from sea water _____

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More differences between compounds and mixtures

1 Complete the table by writing the following statements in the correct columns, to show the differences between a mixture and a compound.

melts over a range of temperature
has a fixed melting point
the proportions of the different elements can be varied
the proportions of the different elements are fixed
can be separated into the elements that make it up
difficult or impossible to separate into the elements that make it up
the properties are different from the elements that make it up
the properties are the same as the elements that make it up

Mixture of elements	**Compound**

2 The table shows the melting points of some metals.

Metal	**Description**	**Melting point (°C)**
copper	pure metal	1083
zinc	pure metal	420
brass	mixture of copper and zinc	
lead	pure metal	327
tin	pure metal	232
solder	mixture of tin and lead	

Why are no melting points given for brass and solder?

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What are rocks made of?

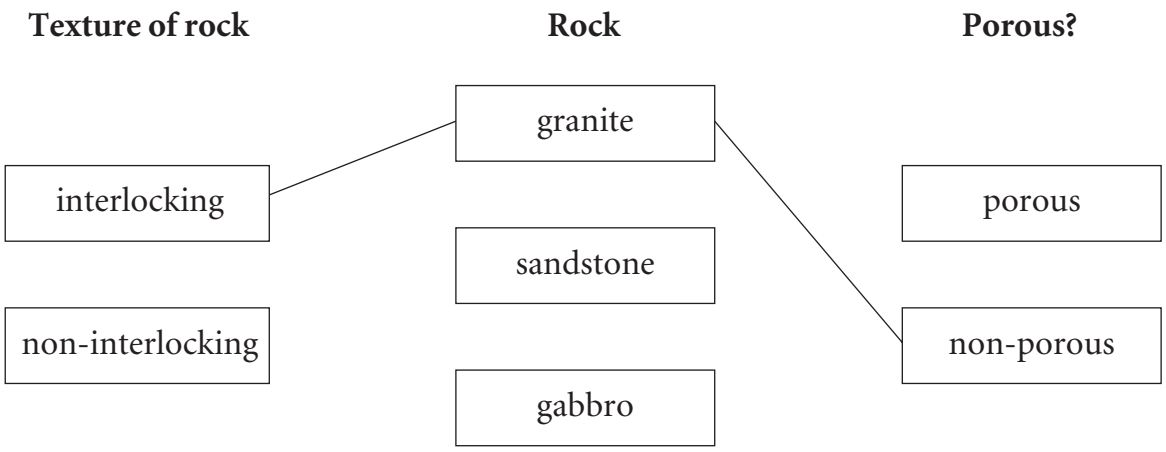
1 Complete the sentences using these words. Use each word once only.

cemented	**crystals**	**grains**	**minerals**

Rocks are made of chemical compounds called _____ that are found as little bits called _____.

In some rocks these bits are glued or _____ together, and in others they have formed _____ that fit together like a jigsaw puzzle.

2 Draw lines to join each rock to the correct boxes. One has been done for you.



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